AP CALCULUS AB - UNIT 1

Guided Notes 1.2: Defining Limits and Using Limit Notation

Name: ____________ **Class:** ____________ **Date:** ____________

Learning Objectives and Essential Question

Essential question: How can we interpret every symbol in finite, one-sided, and infinite limit notation?

- Interpret every symbol in finite, one-sided, and infinite limit notation.
- Translate among words, graphs, tables, and limit statements.
- Use one-sided agreement to determine whether a two-sided limit exists.
- Distinguish a limit, a point value, an infinite limit, and a statement that the limit does not exist.

Key Knowledge, Vocabulary, and Notation

Nearby behavior	$\lim_{x\to c} f(x) = L$ describes $f(x)$ for x close to but not necessarily equal to c.
One-sided notation	$x \to c^-$ means inputs below c; $x \to c^+$ means inputs above c.
Agreement theorem	A finite two-sided limit equals L exactly when both one-sided limits equal L.
DNE	Use DNE when the required finite agreement fails; do not use "undefined" as a substitute.
Infinite limits	$\lim_{x\to c^-} f(x) = +\infty$ means values grow without bound positively from the left; $+\infty$ is not a real output value.

Concept Development and Strategy

1. Read a limit statement in the order: expression, input target, input direction, output destination.
2. Always separate $\lim_{x\to c} f(x)$ from $f(c)$. Either may exist without the other.
3. Before writing a two-sided limit, record the left- and right-hand limits. If they differ, the two-sided limit does not exist.

Worked Example: Words to symbols

Write: "As x approaches -3 from values greater than -3, $p(x)$ decreases without bound."

Answer and Reasoning

$$\lim_{x\to -3^+} p(x) = -\infty.$$

Worked Example: Point value versus limit

A graph approaches 6 from both sides at $x = 2$, has an open circle at $(2, 6)$, and a filled point at $(2, -1)$. State the relevant facts.

Answer and Reasoning

$\lim_{x\to 2^-} f(x) = 6$, $\lim_{x\to 2^+} f(x) = 6$, and therefore $\lim_{x\to 2} f(x) = 6$, while $f(2) = -1$.

Worked Example: Failure by disagreement

Suppose $\lim_{x\to 4^-} h(x) = 2$ and $\lim_{x\to 4^+} h(x) = 7$. What can be concluded?

Answer and Reasoning

The two-sided limit $\lim_{x\to 4} h(x)$ does not exist because the one-sided limits are finite but unequal. No conclusion about $h(4)$ follows.

Guided Practice and Discussion

Directions

Show the mathematical setup, preserve exact values unless a decimal is requested, and justify existence or interpretation statements with the appropriate definition or theorem.

1. Translate $\lim_{x \to 5} f(x) = 9$ into a complete sentence.

2. Write notation for: "$q(x)$ approaches -4 as x approaches 1 from the left."

3. Given $\lim_{x \to 3^-} f(x) = 8$ and $\lim_{x \to 3^+} f(x) = 8$, state the two-sided limit.

4. Given $\lim_{x \to -1^-} g(x) = 0$ and $\lim_{x \to -1^+} g(x) = 5$, state the two-sided conclusion.

5. Suppose $\lim_{x \to a} f(x) = L$. Must $f(a)$ exist? Must L be finite? Answer for the stated notation.

6. Assume $\lim_{x \to c^-} f(x) = L$ and $\lim_{x \to c} f(x) = M$, where both are finite. Prove that $L = M$.

7. Construct a function for which $\lim_{x \to 0^-} f(x) = 2$, $\lim_{x \to 0^+} f(x) = 2$, and $f(0)$ is undefined.

AP Reasoning Check

Multiple-Choice Check

Choose the best answer. Explain why at least one distractor is incorrect during class discussion.

1. If $\lim_{x\to 2^-} f(x) = 5$ and $\lim_{x\to 2^+} f(x) = 5$, which must be true?

(A) $f(2) = 5$

(B) $f(2)$ exists

(C) $\lim_{x\to 2} f(x) = 5$

(D) f is differentiable at 2

2. Which notation means that $g(x)$ decreases without bound as x approaches 4 through values less than 4?

(A) $\lim_{x\to 4^-} g(x) = -\infty$

(B) $\lim_{x\to 4^+} g(x) = -\infty$

(C) $\lim_{x\to -4^-} g(x) = +\infty$

(D) $g(4) = -\infty$

3. A calculator table produces the data below. Which notation best describes the behavior?

x	1.99	1.999	2.001	2.01
$g(x)$	−48.7	−498.7	501.3	51.3

(A) $\lim_{x\to 2} g(x) = 0$

(B) $\lim_{x\to 2^-} g(x) = -\infty$ and $\lim_{x\to 2^+} g(x) = +\infty$

(C) $\lim_{x\to 2} g(x) = +\infty$

(D) $g(2) = 0$

Common Errors and AP Precision

- Keep the value of a function at a point separate from its limiting behavior near that point.
- State one-sided behavior explicitly whenever a two-sided limit or continuity claim depends on agreement from both sides.
- A complete AP justification names the relevant definition, condition, or theorem rather than reporting only a numerical result.

Exit Ticket

Construct a function satisfying $f(0) = 7$, $\lim_{x\to 0^-} f(x) = -1$, and $\lim_{x\to 0^+} f(x) = 3$. State the two-sided limit.